

Current Isolation Agent UI Delivery Timeline

Objective: turn the current Equipment Isolation Agent into a usable review UI for its existing advisory functionality. The UI will expose the agent's current inputs, run progress, P&ID overlays, isolation findings, evidence, warnings, procedure, impact analysis, and audit trace in a clear review workflow.

Estimate: 6 engineering days for one engineer.

Scope

The work reuses the current agent pipeline and its structured result. It does not add new plant-data sources, safety-authorisation workflow, or new isolation-engineering logic.

Included:

- Equipment and project-context selection.
- Agent-run submission, progress, completion, failure, and retry states.
- P&ID image and candidate/boundary overlays.
- Readable presentation of selected isolation candidates, topology findings, coverage, evidence, assurance status, relief findings, instrument context, and downstream impacts.
- LOTO procedure, field holds, warnings, and restoration guidance.
- Agent tool-call trace and the evidence supporting visible findings.
- Saved run results and a usable review of prior runs.
- Responsive desktop and mobile review experience.
- Contract and regression hardening for the existing agent output.

Excluded:

- FHR, PSD, SIC, SAR, isometric, C&E/SIF, SLD, permit-register, or live isolation-register integrations.
- Site-authorised barrier decisions, permit issuance, field execution, authorisation, or lifecycle controls.
- New safety logic beyond presenting current deterministic/agent results clearly and preserving their warnings.

Delivery Timeline (Current UI)

Workstream	Deliverable	Work	Estimate
1. Foundation	Usable run workflow	Create the UI shell; equipment and project-context selection; submit an agent run; show queued, running, complete, and failed states; load a completed run result.	1 day
2. Plan review	Isolation-plan view	Present equipment context, assurance status, warnings, selected candidates, topology paths, coverage, relief findings, evidence, instrument context, and downstream impact in a reviewable hierarchy.	1 day
3. P&ID review	Drawing review	Display the source P&ID, existing candidate/boundary overlays, tag labels, legend, and a details panel linked to result data.	1 day
4. Procedure	Field-oriented procedure view	Present the existing LOTO procedure, isolation order, verification/field holds, warnings-before-steps, restoration guidance, and clear advisory-status language.	0.5 day
5. Evidence and trace	Explainability view	Expose evidence classification, validator rationale, unavailable-analysis reasons, agent audit trace, and source references	0.5 day

Workstream	Deliverable	Work	Estimate
		without exposing raw internal pipeline payloads by default.	
6. Run history	Saved-result access	Link to existing saved result, P&ID, and trace artifacts; no run-management interface beyond current artifacts.	0.5 day
7. Hardening	Release-ready current-agent UI	Handle empty/missing data, agent failures, loading states, responsive layout, output-contract validation, and focused offline regression coverage.	1.5 days
Total	Current-agent UI	A usable advisory review product over all capabilities already produced by the current agent.	6 days

Implementable Requirements Timeline

This is a focused additional increment after the current UI estimate. It improves the current graph-based agent and its review output only. It excludes lifecycle/authorisation, offline field execution, emergency mode, permits, canonical exports, multi-tenancy, and every requirement blocked by unavailable plant data or upstream-platform dependencies.

Workstream	Requirements	Deliverable	Estimate
Operational review language	REQ-OM-01, REQ-IN-12	Use field-oriented action/proving wording and add the electrical-out-of-scope powered-device checklist.	2–3 days
Scope intake	REQ-IN-15..16	Capture the missing structured work-scope fields needed by the current review and warning output.	2–3 days
Topology and evidence review	REQ-F1-02, REQ-D-03, REQ-D-07	Expand current direction-independent graph traversal, show candidate series relationships, and add the valve-integrity limitation statement.	2–3 days
Procedure and explainability	REQ-F6-01, REQ-F6-04, REQ-F8-03	Add structured per-step fields, generic hold-point presentation, and clear confidence categories to the current advisory output.	2–3 days
Regression hardening	REQ-SA-02, REQ-SA-05	Add focused requirements-to-test mapping and offline regression cases for the current-agent success, gap, and failure paths.	2–3 days
Total	Focused implementable increment	Additional work after the six-day current UI increment; excludes blocked data, operational workflow, and new product modes.	10–15 days

Requirements Feasibility

Requirement	Feasibility	Assessment
REQ-OM-01: Operational language	Implementable	Outputs include tags and generic procedure language; add consistent field location, action, and proving terminology.
REQ-OM-02: Offline tablet execution	Implementable	Requires a field execution UI, offline storage, synchronisation, and a glove-oriented tablet workflow.
REQ-OM-03: Plan lifecycle	Implementable	Requires a controlled lifecycle and immutable audit trail for plan state changes and edits.
REQ-OM-04: Authorisation record	Implementable	Requires a human authoriser, approval timestamp, and approved plan/graph/rule-version record.
REQ-OM-05: Re-authorisation after change	Implementable	Requires change detection, safety-significance diff, invalidation, and re-authorisation workflow.
REQ-OM-06: Shift handover	Implementable	Requires point-level execution state and a handover view.
REQ-ARCH-01: Deterministic safety core	Implementable	Candidate selection, validation, and LOTO phases are deterministic; add HSC/RBC derivation, exhaustive traversal, barrier rules, and sequencing core.
REQ-ARCH-02: LLM confined to periphery	Implementable	The deterministic validator remains authoritative; complete the verified safety core before confining the LLM to the specified peripheral role.
REQ-ARCH-03: Verified LLM plan content	Implementable	Requires a complete fired-rule trace and a visual distinction for generated narrative.
REQ-IN-01: Version-stamped UniGraph	Implementable with UniGraph support	Requires pinning and recording an authoritative UniGraph graph version.
REQ-IN-02: Unknown fluid	Implementable	Requires the HSC-4 fallback, gap record, and non-authorisable status for an unmapped service code.
REQ-IN-03: FHR revision	Not implementable with current inputs	No controlled FHR or FHR revision is available to record on a plan.
REQ-IN-04: PSD assumptions	Not implementable with current inputs	No PSD declarations exist to reproduce on a certificate with the declarer's identity.
REQ-IN-05: Incomplete PSD defaults	Implementable	Requires conservative PSD defaults and explicit provenance.
REQ-IN-06: PSD validity	Not implementable with current inputs	No PSD validity record or authorisation gate exists.
REQ-IN-07: SIC derivation trace	Implementable with SIC	Requires a versioned SIC and parameter-level derivation trace.

Requirement	Feasibility	Assessment
REQ-IN-08: Default SIC	Implementable	Generic conservative policy exists; make it a controlled OSHA-aligned SIC profile.
REQ-IN-09: SAR hot-work gate	Not implementable with current inputs	No SAR supports the required hot-work permit gate.
REQ-IN-10: SAR effort/access	Not implementable with current inputs	No SAR supports effort estimation or emergency accessibility.
REQ-IN-11: Isometric geometry	Blocked by dependency	Drawing and Document AI extraction of geometry is not available in UniGraph.
REQ-IN-12: Electrical-out-of-scope placeholder	Implementable	Requires a standing electrical limitation section and powered-device checklist.
REQ-IN-13: Electrical isolation register	Not implementable with current inputs	No electrical isolation register input or sequence integration exists.
REQ-IN-14: SLD electrical hierarchy	Blocked by dependency	No Convert-for-SLD hierarchy is available.
REQ-IN-15: Work-scope inputs	Implementable	Equipment-tag scope is supported; add natural language, CMMS, graphical, and named-system intake.
REQ-IN-16: Work-scope details	Implementable	Limited scope flags exist; add activity, duration, shift, containment, entry, hot-work, and off-site fields.
REQ-D-01: RBC matrix overrides	Implementable with SIC	Requires a versioned SIC matrix, override justification, approver, and derivation trace.
REQ-D-02: Small-bore rule	Not implementable with current inputs	Nominal diameter and a controlled small-bore threshold are unavailable.
REQ-D-03: Series barrier verification	Implementable	HILT topology can identify candidate valve connectivity; add complete per-path barrier verification.
REQ-D-04: Valid bleed	Not implementable with current inputs	This requires per-path barriers, bleed topology, and PSD destination state.
REQ-D-05: No-bleed escalation	Not implementable with current inputs	Outward topology search is feasible, but a compliant result requires RBC, bleed, and flanged-joint inputs.
REQ-D-06: Proving method	Not implementable with current inputs	SIC proving criteria and validated selected barriers are unavailable.

Requirement	Feasibility	Assessment
REQ-D-07: Valve-integrity limitation	Implementable	Current outputs include warnings and field holds; add the required standing limitation statement on every plan.
REQ-D-08: Failed-proving re-derivation	Implementable	Requires field proving capture, plan supersession, and re-authorisation workflow.
REQ-D-09: Derivation trace	Not implementable with current inputs	No HSC/EC/RBC derivation exists to trace.
REQ-F1-01: Boundary to RBC	Not implementable with current inputs	Traversal exists in limited form, but cannot terminate at a required barrier configuration until HSC/RBC are available.
REQ-F1-02: Direction-independent traversal	Implementable	Current traversal is bounded/local; expand it to traverse all authoritative UniGraph paths regardless of direction.
REQ-F1-03: Per-path HSC/RBC	Not implementable with current inputs	FHR, PSD, SIC, device attributes, and complete scope are absent.
REQ-F1-04: Nested boundaries	Not implementable with current inputs	A PSD with active boundary points and HSC/RBC inputs is absent.
REQ-F1-05: Shared isolation points	Not implementable with current inputs	A PSD active-isolation register with dependency/ownership data is absent.
REQ-F1-06: Package-unit boundaries	Blocked by dependency	Current UniGraph code does not model package units or declared battery-limit connections.
REQ-F1-07: Non-isolatable paths	Not implementable with current inputs	The reporting UI is possible, but the required no-valid-RBC conclusion is not.
REQ-F2-01: Residual and stored energy	Not implementable with current inputs	Isometrics, volumes, elevation, thermal/chemical service data, and mechanical design data are absent.
REQ-F2-02: Energy mitigation	Not implementable with current inputs	Energy inventory, vent/drain/purge topology, and PSD destination state are absent.
REQ-F2-03: Undrained low points	Not implementable with current inputs	Isometric low-point and drain-location data are absent.
REQ-F2-04: Mechanical energy	Not implementable	Equipment-specific rotation, spring, accumulator, and tension data are absent.

Requirement	Feasibility	Assessment
	with current inputs	
REQ-F3-01: Instrument impulse isolation	Not implementable with current inputs	Root-valve/manifold topology and DP sequence data are absent.
REQ-F3-02: Control consequence	Not implementable with current inputs	Loop, final-element, interlock, and fail-action data are absent.
REQ-F3-03: C&E overrides	Blocked by dependency	Queryable C&E data from the Drawing and Document AI Agent is unavailable.
REQ-F3-04: SIF voting/inhibition	Not implementable with current inputs	SIF voting, C&E, deviation workflow, and SIC duration policy are absent.
REQ-F3-05: Utility-dependent final elements	Not implementable with current inputs	Utility dependencies, final-element fail positions, and utility state are absent.
REQ-F3-06: Safety-critical impairments	Not implementable with current inputs	Safety-system relationships and an impairment workflow are absent.
REQ-F4-01: Depressurisation	Not implementable with current inputs	Vent topology, PSD destination state, SIC targets, and a valid derived boundary are absent.
REQ-F4-02: Drain-down	Not implementable with current inputs	Drain topology, volumes, elevations, low points, and PSD destination state are absent.
REQ-F4-03: Purge/inerting	Not implementable with current inputs	SIC purge rules, connection topology, volumes, targets, and test points are absent.
REQ-F4-04: Nitrogen/asphyxiation	Not implementable with current inputs	A generic warning is possible, but a derived nitrogen plan and entry context are absent.
REQ-F4-05: Gas tests	Not implementable with current inputs	Test-point topology, SIC criteria, repeat intervals, and permit context are absent.
REQ-F5-01: Live-plant impact	Not implementable	HILT reachability flags possible impacts, but a complete derived boundary and live plant state are absent.

Requirement	Feasibility	Assessment
	with current inputs	
REQ-F5-02: Relief integrity	Not implementable with current inputs	Relief relationships, PSD state, CSO/LO state, and deviation workflow are absent.
REQ-F5-03: Availability	Not implementable with current inputs	PSD availability and train/fire-pump/detector coverage data are absent.
REQ-F5-04: Plant-mode viability	Not implementable with current inputs	PSD plant mode and mode-specific service requirements are absent.
REQ-F5-05: SIMOPS	Not implementable with current inputs	Active isolation and permit records with scope, location, status, and timing are absent.
REQ-F6-01: Numbered sequence	Implementable	Generic LOTO steps exist; add per-device actions, end positions, roles, proving methods, and dependencies.
REQ-F6-02: Safe sequence	Not implementable with current inputs	The fixed LOTO phase order is present; derived process, joint, drain/purge, blind, and energy dependencies are absent.
REQ-F6-03: Reinstatement	Not implementable with current inputs	Blind IDs, override records, normal positions, test requirements, and execution state are unavailable.
REQ-F6-04: Signed hold points	Implementable	Generic holds can be displayed; add applicable-hold derivation and sign-off workflow.
REQ-F6-05: Partial re-isolation	Implementable	Requires a point-level lifecycle and staged-work model.
REQ-F7-01: Isolation schedule	Not implementable with current inputs	Current output has candidates/procedure, but location, lock/tag, discipline, and execution data are absent.
REQ-F7-02: Marked-up P&ID	Not implementable with current inputs	Candidate overlays exist, but a complete derived boundary/blind/bleed/drain/vent markup requires unavailable plan inputs.
REQ-F7-03: Bill of materials	Not implementable with current inputs	Piping class, geometry, blind/gasket data, and complete derivation are absent.
REQ-F7-04: Certificate draft	Not implementable	A site template and complete controlled plan data are absent.

Requirement	Feasibility	Assessment
	with current inputs	
REQ-F7-05: Assumption/gap/deviation register	Implementable with PSD	Warnings and gaps exist; add a unified PSD/deviation/impairment/override register.
REQ-F7-06: Effort estimate	Not implementable with current inputs	SAR, blind data, and duration/discipline data are absent.
REQ-F7-07: Canonical export	Implementable	Requires a documented canonical JSON/XML plan schema and CMMS adapters.
REQ-F8-01: Point derivation	Not implementable with current inputs	Evidence and traces exist, but HSC/EC/RBC inputs needed for a complete point derivation are absent.
REQ-F8-02: Interrogatives	Implementable with required inputs	Requires a what-if and complete path-explanation interface backed by versioned inputs.
REQ-F8-03: Per-point confidence	Implementable	Requires categories separating graph, rule, declared, and missing-data conclusions.
REQ-F8-04: Byte-identical reproducibility	Implementable with version pinning	Requires all inputs to be version-pinned and a deterministic planning path.
REQ-E-01: Emergency advisory position	Implementable	Requires an emergency-mode output and standing emergency advisory statement.
REQ-E-02: Emergency enablement	Implementable	Requires site-level emergency-mode configuration.
REQ-E-03: Emergency intake	Implementable	Requires an emergency event input and natural-language emergency parsing.
REQ-E-04: Emergency PSD fallback	Implementable	Requires emergency-mode PSD handling and an all-live fallback.
REQ-E-05: Flow-limiting actions	Implementable with UniGraph data	Requires emergency release-source traversal and action-ranking logic.
REQ-E-06: Remote isolation	Implementable with UniGraph data	Requires a remote-device action model and emergency prioritisation.
REQ-E-07: Residual inventory	Blocked by dependency	Line and equipment volume data are absent.
REQ-E-08: Blowdown route	Not implementable with current inputs	Requires blowdown topology and destination state.
REQ-E-09: Manual accessibility	Blocked by dependency	SAR accessibility data is unavailable.

Requirement	Feasibility	Assessment
REQ-E-10: Escalation targets	Blocked by dependency	SAR proximity and inventory data are unavailable.
REQ-E-11: Plant consequences	Implementable with required inputs	Requires emergency action impact analysis and plant-state context.
REQ-E-12: Degraded emergency output	Implementable	Requires an emergency-mode output path that reports assumptions and gaps.
REQ-E-13: Progressive response	Implementable	Requires an emergency response pipeline and performance implementation.
REQ-E-14: Emergency logging	Implementable	Requires a retained emergency output record.
REQ-E-15: Emergency non-goals	Implementable	Requires these constraints to be enforced by the emergency-mode workflow.
REQ-P-01: Permit types	Implementable with SIC	Requires permit-generation capability and SIC permit configuration.
REQ-P-02: Permit/isolation link	Implementable	Requires a permit model linked to a controlled isolation plan.
REQ-P-03: Permit content	Not implementable with current inputs	FHR, proving, gas-test, SAR, SIMOPS, and template data are absent.
REQ-P-04: Permit validity	Implementable	Requires a permit lifecycle and proving-validity model.
REQ-P-05: Permit conflicts	Not implementable with current inputs	SAR and active-permit data are absent.
REQ-P-06: SAR hot-work gate	Not implementable with current inputs	No SAR exists.
REQ-P-07: Permit export	Implementable	Requires a canonical schema and permit adapter.
REQ-P-08: Permit draft only	Implementable	Requires a permit workflow that preserves authorisation as a human action.
REQ-SA-01: Classification argument	Implementable with client input	Requires an agreed client safety-tool classification and validation strategy.
REQ-SA-02: Assurance and traceability	Implementable	Some deterministic code and unit tests exist; add full coverage and a requirements-to-test matrix.
REQ-SA-03: Version pinning	Implementable	Requires authorised plans to record the required input and model versions.
REQ-SA-04: Change control	Implementable	Requires formal MoC and release gates for derivation and policy changes.

Requirement	Feasibility	Assessment
REQ-SA-05: Scenario regression	Implementable	Requires the specification scenario catalogue to become an executable regression suite.
REQ-SA-06: Independent review	Implementable with process-safety input	Requires a recorded independent process-safety review.
REQ-SA-07: Failure controls	Implementable	Warnings and deterministic validation address some failure modes; add the full specified controls.
REQ-SA-08: Limitations statement	Implementable	Current outputs carry safety warnings; add the required standing limitations statement.
REQ-NF-01: Performance	Unverified	No representative tests demonstrate the specified M1/M2/M3 performance targets.
REQ-NF-02: Scale	Unverified	No tests demonstrate 500 P&IDs, 500,000 nodes, or 500 concurrent active isolations.
REQ-NF-03: Immutable auditability	Implementable	Agent traces exist; add an immutable audit trail across inputs, edits, authorisation, and retention.
REQ-NF-04: Multi-tenancy	Implementable	Requires tenant segregation and per-tenant controlled configuration.
REQ-NF-05: Offline field operation	Implementable	Requires offline execution and synchronisation.
REQ-NF-06: Localisation	Implementable	Requires configurable units and template translation.
REQ-DEP-01: Graph-to-drawing links	Implementable with Convert P&ID	HILT/STLM bounding boxes provide overlay coordinates; full bi-directional entity/drawing coordinate links need verification.
REQ-DEP-02: Valve/joint attributes	Implementable with Convert P&ID	Some component classes exist; the complete valve, position, blind, and joint attribute set must be provided.
REQ-DEP-03: UniGraph versioning	Blocked by dependency	Authoritative graph version/change tracking is not currently consumed by the agent.
REQ-DEP-04: Package battery limits	Blocked by dependency	The current UniGraph backend/parser has no package-unit or battery-limit model.
REQ-DEP-05: Queryable C&E	Blocked by dependency	Drawing and Document AI C&E extraction is not integrated.
REQ-DEP-06: Isometric geometry	Blocked by dependency	Drawing and Document AI isometric extraction is not integrated.
REQ-DEP-07: Loop-drawing extraction	Blocked by dependency	No loop-drawing power/signal extraction is available.
REQ-DEP-08: SLD hierarchy	Blocked by dependency	Convert for SLD is explicitly future work.
REQ-DEP-09: Canonical export/adapters	Implementable	Requires a canonical export schema and Maximo/SAP adapters.

Question

Do we need to build the PSD workflow, or will another operational system provide PSD snapshots? If we need to build it, who owns the declared plant-state data and approvals?

Definition of Done

- A user can select an equipment tag and explicit project context, start the current agent, and follow the run to completion or a clear failure state.
- A completed result is reviewable without reading raw JSON or an HTML artifact directly.
- The user can inspect the P&ID overlay and connect a selected device or warning to its evidence and rationale.
- Procedure, warnings, assurance status, field holds, relief findings, instrument context, downstream impact, and agent trace are visible in the UI.
- Prior results remain accessible for review.
- Missing source information is shown as unavailable or requiring field review; it is never presented as a validated safety conclusion.
- The existing offline test suite and UI/API contract checks pass for normal, incomplete-data, and failed-run cases.

Assumptions

- The current API can invoke `agent.runner.run_agent_pipeline` and persist result artifacts as described in the repository.
- Existing P&ID images, overlay data, structured output, and agent traces remain available to the UI.
- Estimates exclude stakeholder review, deployment infrastructure, design-system creation, external-data onboarding, and formal process-safety validation.